Determinants

Section 1: Cofactor Expansion

1 Evaluate the determinant of $A = [[2, 1, 0], [3, -2, 4], [1, 0, 5]]$.

a. -38

b. 38

c. -22

d. 14

2 What is the determinant of any upper triangular or lower triangular matrix?

a. The product of the diagonal entries

b. The sum of the diagonal entries

c. Always 1

d. Always 0

3 If a row of matrix A consists entirely of zeros, what is $\det(A)$?

a. 0

b. 1

c. Undefined

d. -1

Section 2: Row Operations and Determinants

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4 How does swapping two rows of a square matrix affect its determinant?

a. The determinant is multiplied by -1

b. The determinant remains unchanged

c. The determinant is squared

d. The determinant becomes 0

5 How does adding a multiple of one row to another row affect the determinant?

a. The determinant does not change

b. The determinant is multiplied by the multiple

c. The determinant is negated

d. The determinant is divided by the multiple

6 If one row of an $n \times n$ matrix is multiplied by scalar c, what happens to $\det(A)$?

a. The determinant is multiplied by c

b. The determinant is multiplied by c^n

c. The determinant is unchanged

d. The determinant is multiplied by c^2

Section 3: Determinant Properties & Multiplicative Rule

7 For any $n \times n$ matrices A and B, what is $\det(AB)$?

a. $\det(A) \cdot \det(B)$

b. $\det(A) + \det(B)$

c. $\det(BA) + 1$

d. $(\det A)^{\det B}$

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8 What is $\det(A^T)$ in relation to $\det(A)$?

a. $\det(A^T) = \det(A)$

b. $\det(A^T) = -\det(A)$

c. $\det(A^T) = 1/\det(A)$

d. $\det(A^T) = (\det A)^2$

9 Under what condition is $\det(A) = 0$?

a. If and only if A is singular (not invertible)

b. If and only if all entries are zero

c. Whenever A is symmetric

d. Whenever A has positive trace

Section 4: Cramer's Rule and Geometric Interpretations

10 What geometric quantity does $|\det(A)|$ represent for a 2×2 matrix with columns $\mathbf{a}_1, \mathbf{a}_2$?

a. The area of the parallelogram formed by $\mathbf{a}_1$ and $\mathbf{a}_2$

b. The perimeter of the parallelogram

c. The distance between $\mathbf{a}_1$ and $\mathbf{a}_2$

d. The angle between the columns

11 What does $|\det(A)|$ represent for a 3×3 matrix?

a. The volume of the parallelepiped determined by the column vectors

b. The surface area of the parallelepiped

c. The length of the body diagonal

d. The sum of the lengths of the columns

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12 According to Cramer's Rule for solving $A\mathbf{x} = \mathbf{b}$ where $\det(A) \neq 0$, the ith variable is given by:

a. $x_i = \frac{\det(A_i(\mathbf{b}))}{\det(A)}$

b. $x_i = \frac{\det(A)}{\det(A_i(\mathbf{b}))}$

c. $x_i = \det(A) \cdot b_i$

d. $x_i = \frac{b_i}{\det(A)}$

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